Introduction to LLM Agents

June 1st, 2026

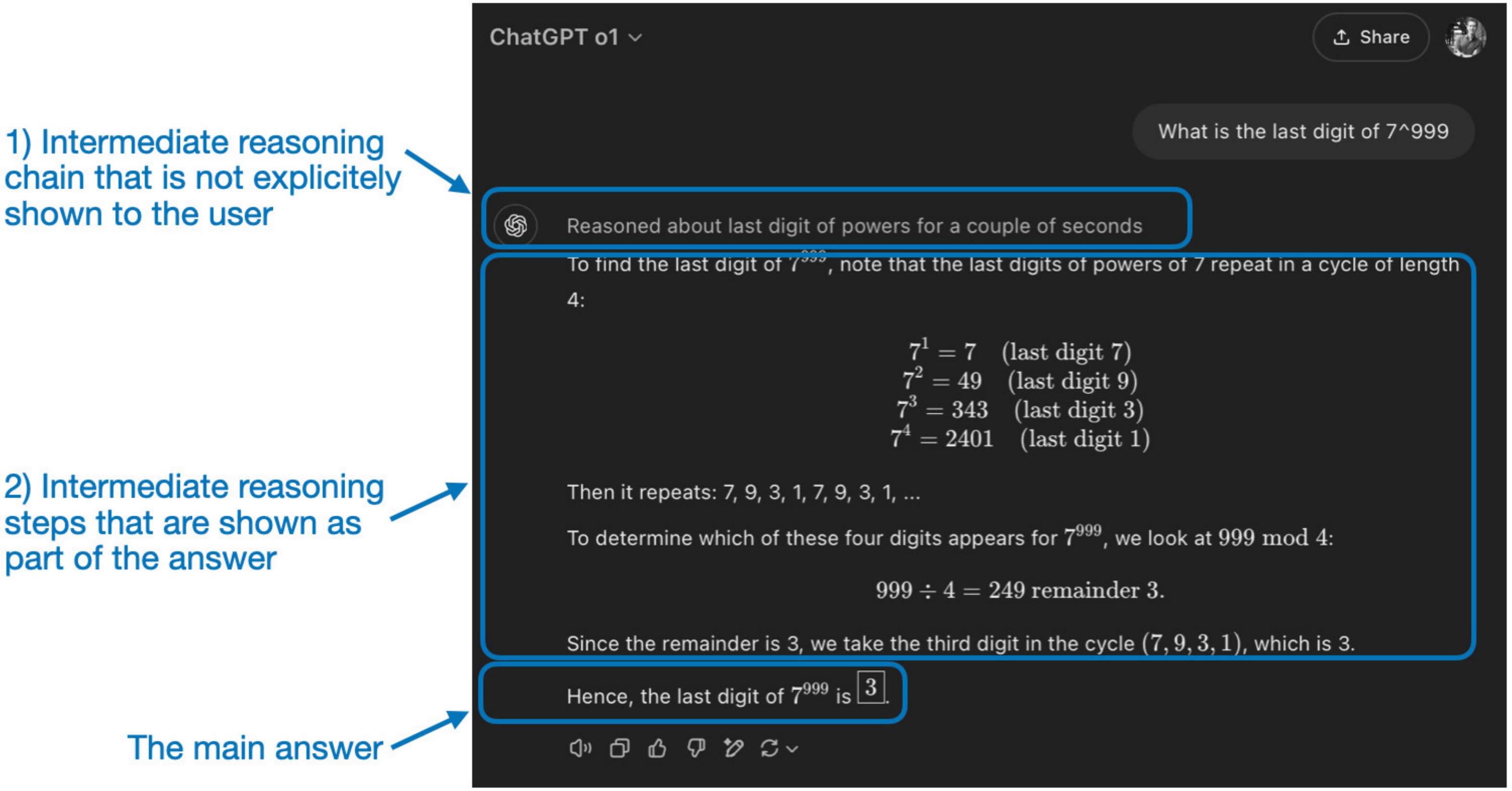
Reasoning

LLMs are great at most tasks, but not all: multi-step planning, arithmetic, complex code...

Reasoning

Solution: reasoning

- Is an emergent capability, but annotated data used to improve it



[Raschka]

When to Use Reasoning

Good at	Bad at
+ Deductive or inductive reasoning (e.g., riddles, math proofs)	– Fast and cheap responses (more inference time)
+ Chain-of-thought reasoning (breaking down multi-step problems)	– Knowledge-based tasks (hallucination)
+ Complex decision-making tasks	– Simple tasks (“overthinking”)
+ Better generalization to novel problems	

[Raschka]

How to Train a Reasoning Model

Pretraining → SFT* → RLHF**

The first "reasoning model" was introduced by OpenAI's o1-preview in September 2024, followed by Qwen with Questions and Gemini 2.0 Flash Experimental

* Supervised Fine-Tuning

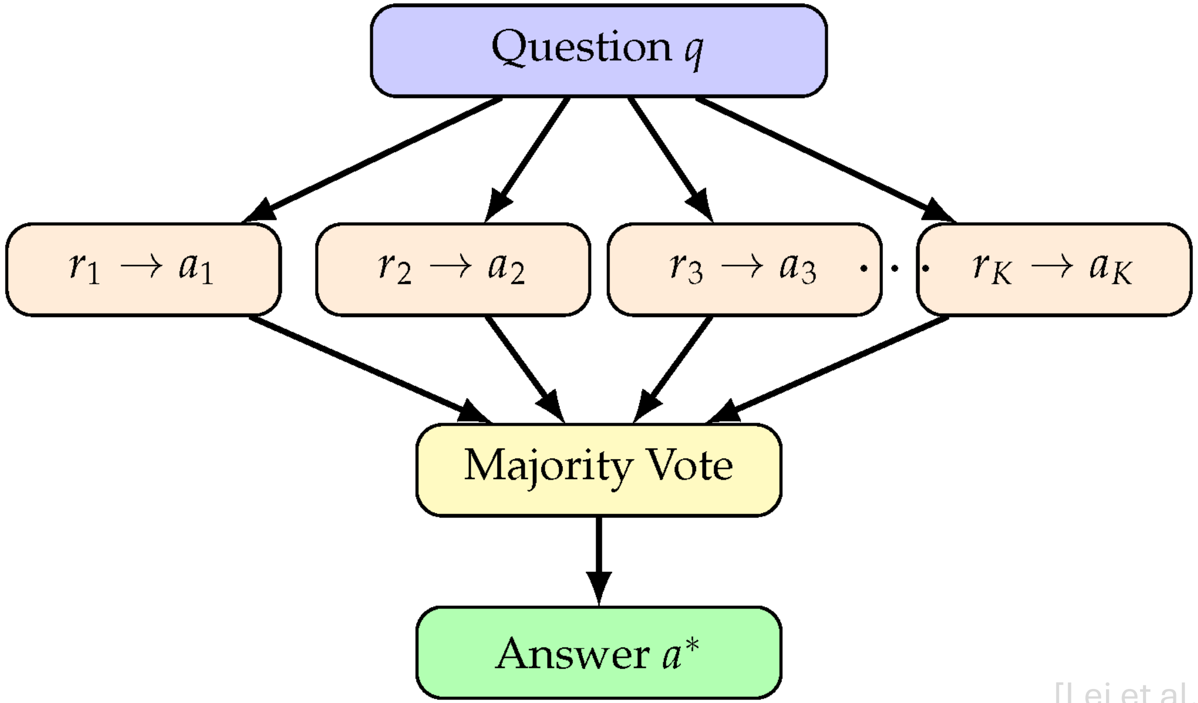
** Reinforcement Learning with Human Feedback

How to Improve Reasoning

- Inference-time scaling

How to Improve Reasoning

- **Inference-time scaling:** increasing computational resources during inference
  - Self Consistency aka Majority Voting
  - Chain-of-Thought prompting



[Lei et al.]

Chain-of-Thought

- **Key idea:** improve performance by verbalising the thinking process
- One highly complex prediction → multiple simple predictions
- Adaptive: more complex question → more reasoning steps

Regular prompting

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?
A: The answer (arabic numerals) is

(Output) *8* ✗

Chain-of-thought prompting

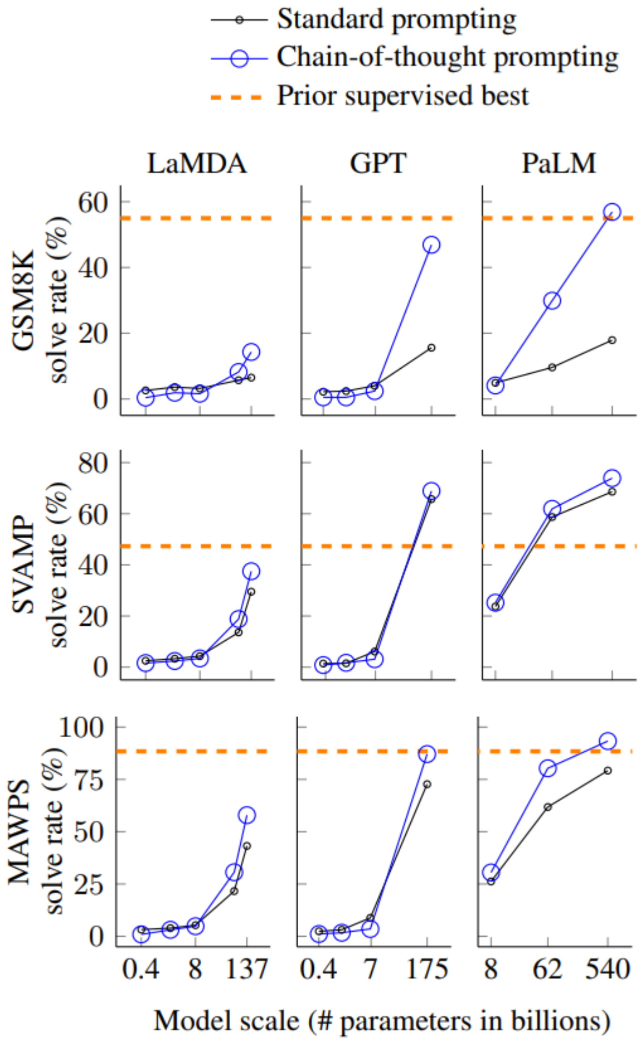
Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?
A: ***Let's think step by step.***

(Output) *There are 16 balls in total. Half of the balls are golf balls. That means that there are 8 golf balls. Half of the golf balls are blue. That means that there are 4 blue golf balls.* ✓

[Kojima et al.]

Chain-of-Thought

- CoT performance scales with model size



[Wei et al.]

Chain-of-Thought

No.	Category	Template	Accuracy
1	instructive	Let's think step by step.	**78.7**
2		First,	77.3
3		Let's think about this logically.	74.5
4		Let's solve this problem by splitting it into steps.	72.2
5		Let's be realistic and think step by step.	70.8
6		Let's think like a detective step by step.	70.3
7		Let's think	57.5
8		Before we dive into the answer,	55.7
9		The answer is after the proof.	45.7
10	misleading	Don't think. Just feel.	18.8
11		Let's think step by step but reach an incorrect answer.	18.7
12		Let's count the number of "a" in the question.	16.7
13		By using the fact that the earth is round,	9.3
14	irrelevant	By the way, I found a good restaurant nearby.	17.5
15		Abrakadabra!	15.5
16		It's a beautiful day.	13.1
-		(Zero-shot)	17.7

- **Key point:** different ways to elicit CoT yield different performance

[Kojima et al.]

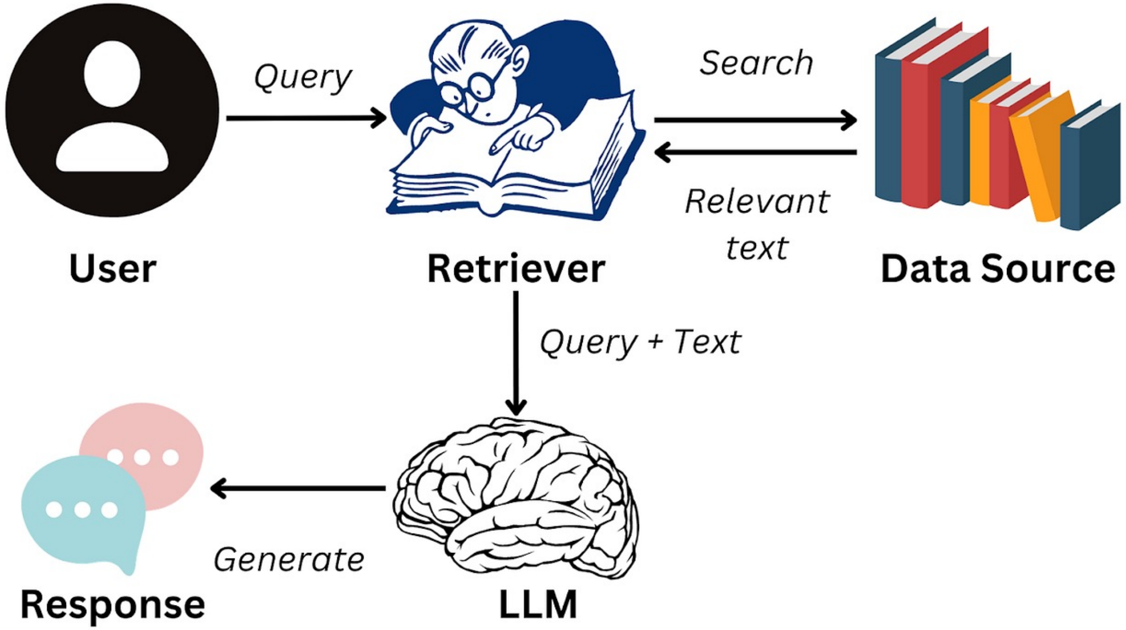
Challenges of LLMs

- Can’t say “I don’t know”
- Out-of-date/ generic information
- Terminology confusion
- Responses from unreliable sources

$\rightarrow$ need pre-determined, clean, reliable sources $\rightarrow$ RAG

RAG

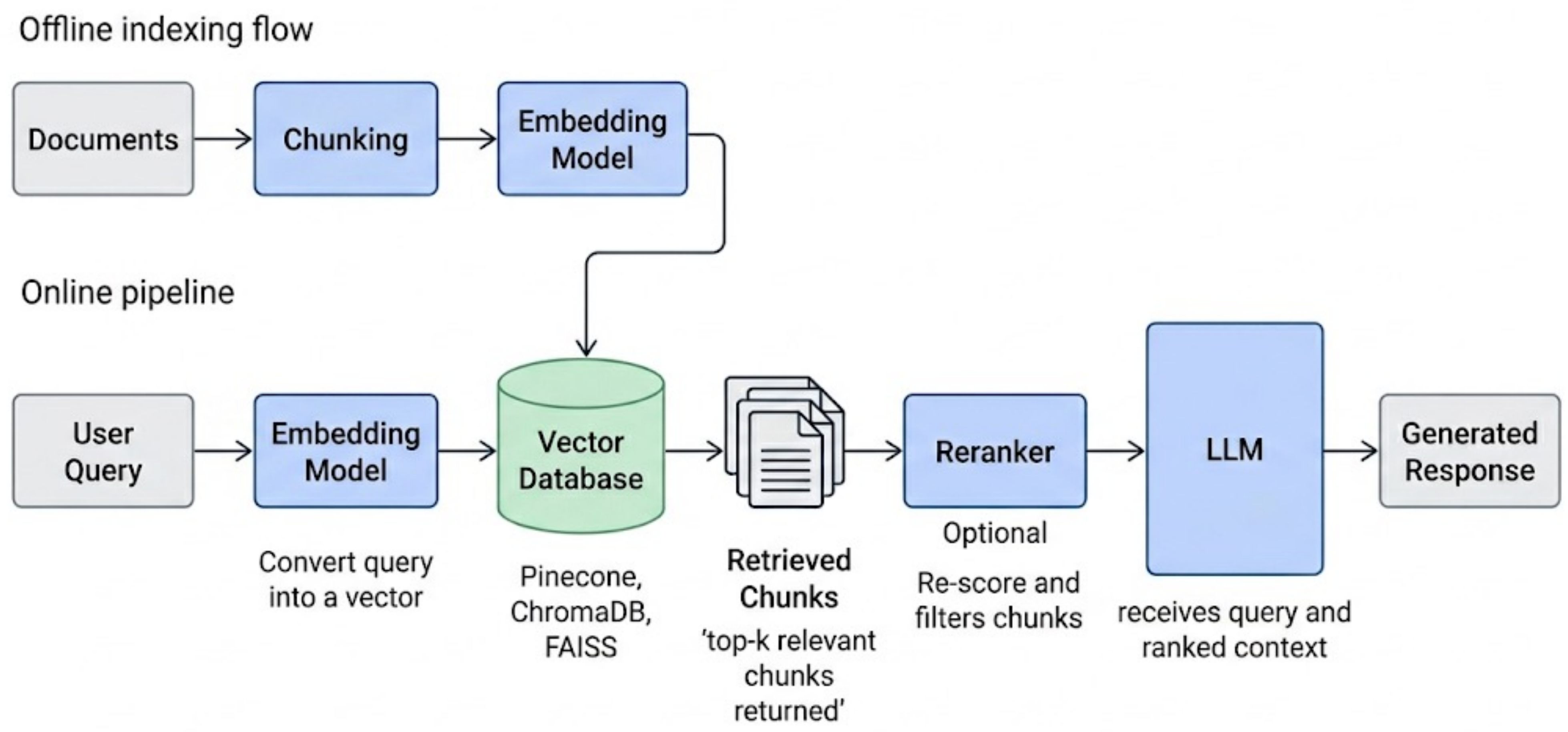
- **Key idea:** enhance LLM by connecting it to an external data source
- Retrieve facts via search and integrate them into the output



RAG Components

- **Knowledge base:** external data repository. Chunking and embedding
- **Retriever:** embedding model and Approximate Nearest Neighbour Search
- **Integration layer:** combine user query with the retrieved information
- **Generator:** the LLM

RAG Components



Why Use RAG?

- Mitigates hallucinations
- Gives access to relevant, specialized, up-to-date data
- Builds user trust
- More developer control
- Cost-efficient implementation and scaling
- Greater data security

Exercise

From RAG to Agents

After CoT and RAG, what can an LLM still not do?

Find three recent papers on LLM Agents and tell me which one is most relevant for our course.

From RAG to Agents

After CoT and RAG, what can an LLM still not do?

Find three recent papers on LLM Agents and tell me which one is most relevant for our course.

Model may need to…

- Search the web
- Open several sources
- Decide which sources are relevant
- Compare different sources
- Maybe search again with better keywords

From RAG to Agents

After CoT and RAG, what can an LLM still not do?

Find a train from Bochum to Berlin tomorrow morning, under 80EUR, arriving before noon.

From RAG to Agents

After CoT and RAG, what can an LLM still not do?

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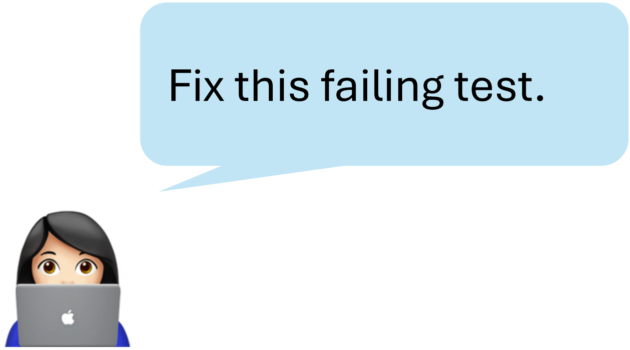
Model may need to…

- Query a live source
- Filter options
- Compare constraints
- Revise the search
- Ask before booking

From RAG to Agents

After CoT and RAG, what can an LLM still not do?



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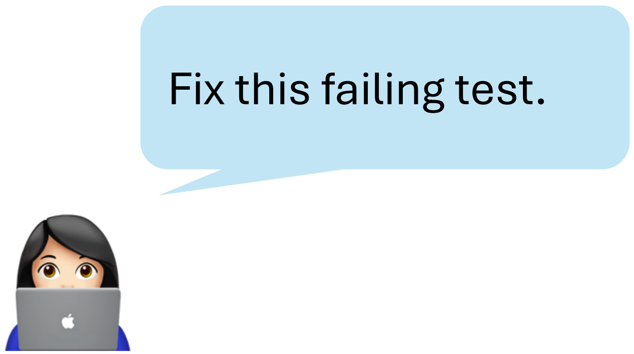

Model may need to…

- Inspect files
- Run tests
- Observe errors
- Edit code
- Run tests again
- Stop when tests pass

From RAG to Agents

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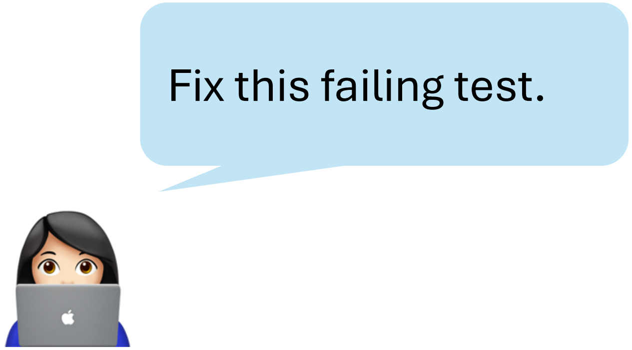

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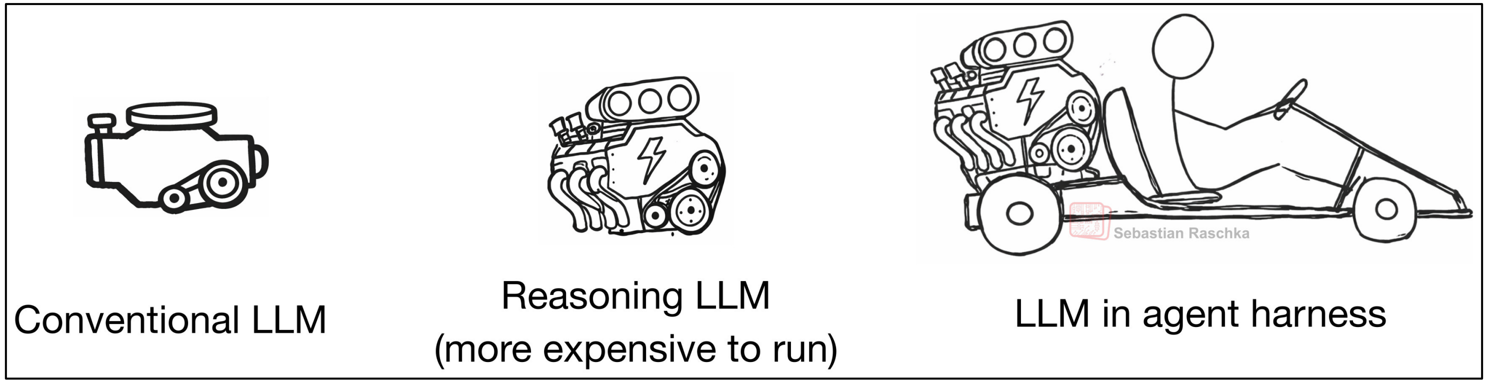
Answer Generation ≠ Task Completion

Three Levels



Plain LLM — **Prompt → Answer**

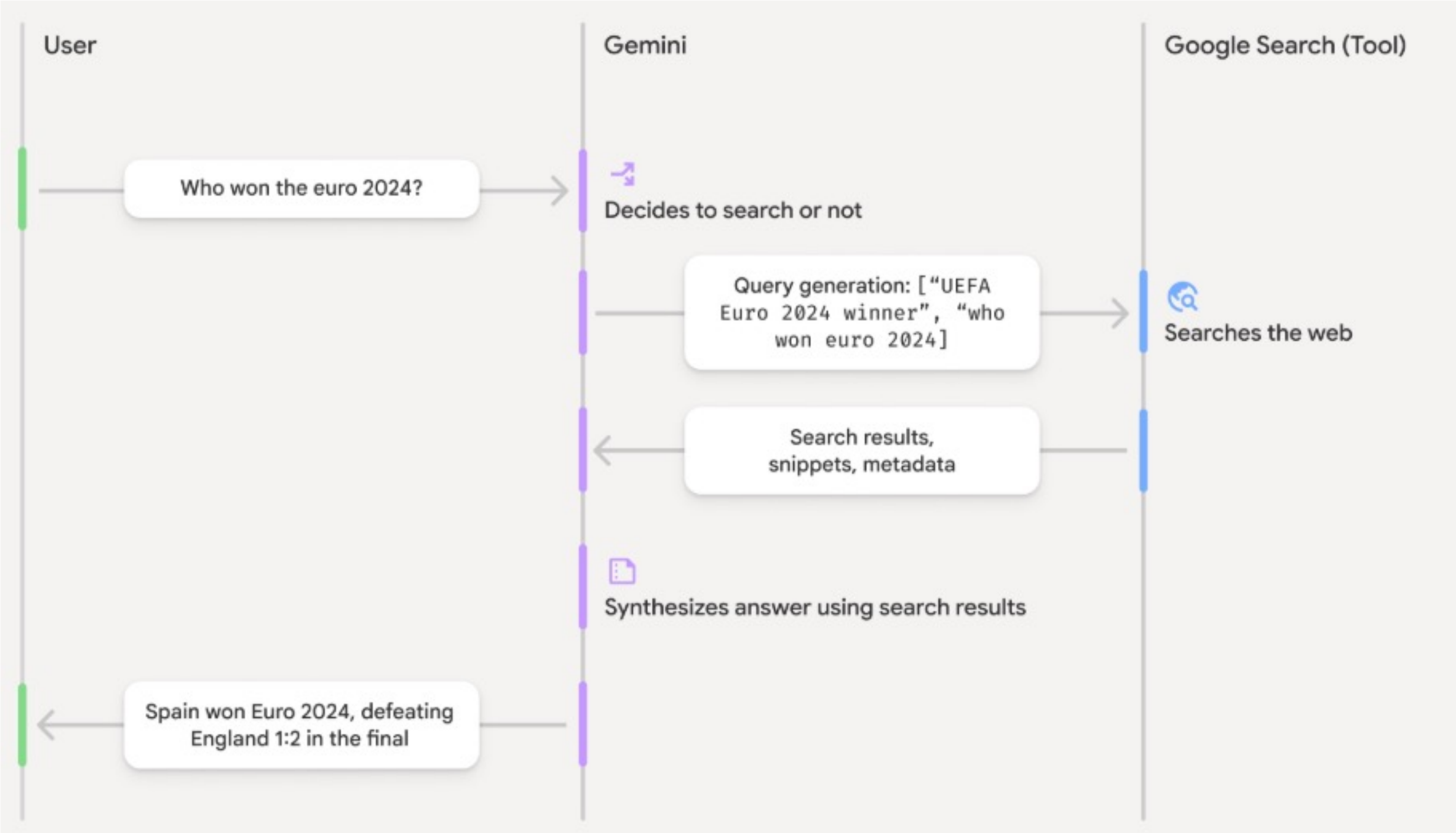
Reasoning LLM — **Query → Think → Answer**

Agentic System — **Goal → Act → Observe → Adapt**

From RAG to Agents

Agents:
LLMs with access to tools and the freedom to use them to get things done.

LLMs and Search



Grounding with Google Search

Google as Another Database

RAG database	**Web search**
Fixed corpus	Open web
Retrieve embedded chunks	Retrieve ranked pages/snippets
Usually private/domain-specific	Usually public/current
Controlled and curated	Noisy
Usually designed for a single retrieval step	Allows for more iterations over queries

Many LLMs now support Web Search

AI Overview

Gemini

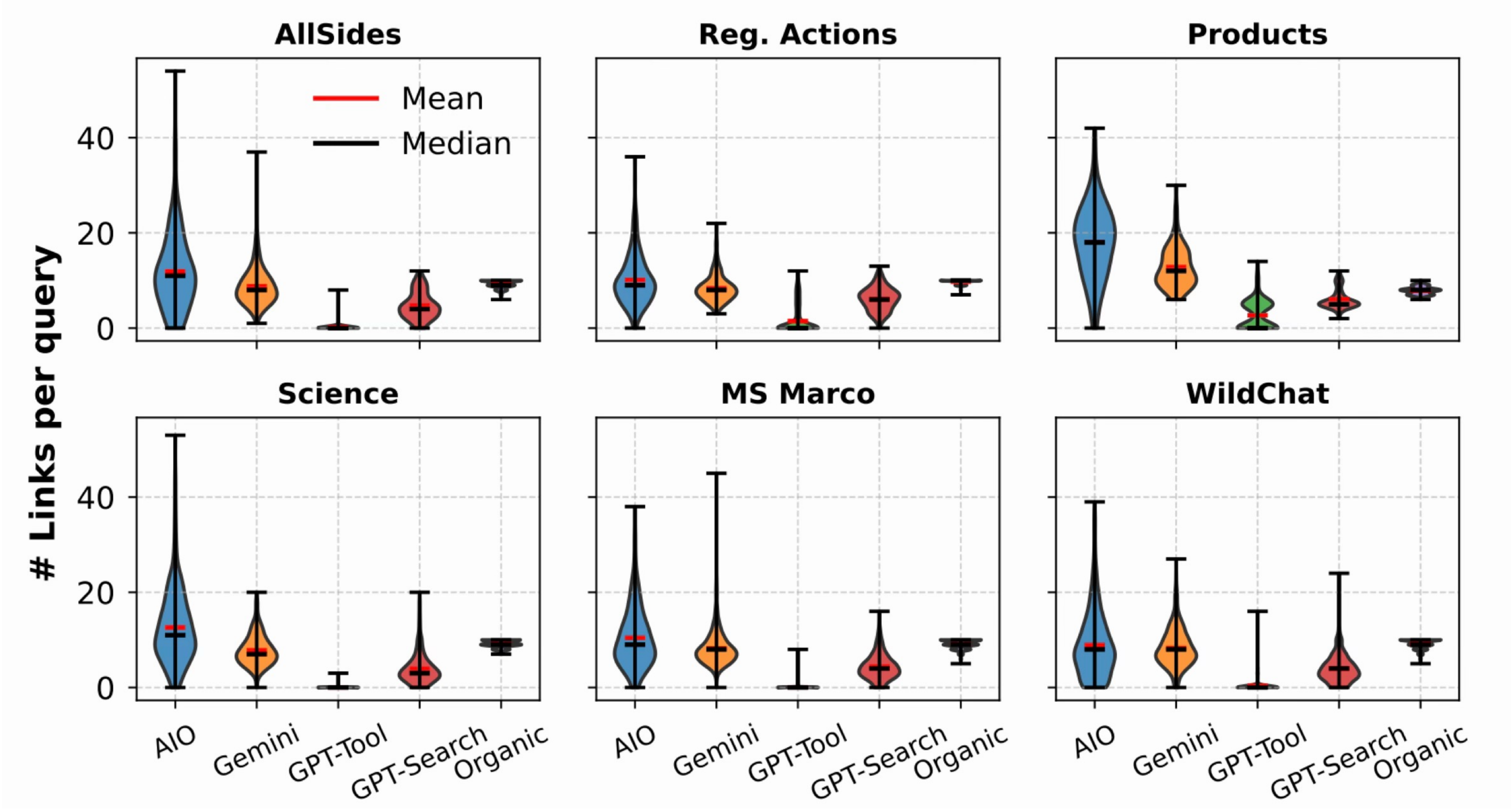
perplexity

OpenAI

v.s. Google

“Characterizing Web Search in The Age of Generative AI”

Internal vs. External Knowledge



Models rely on external information to varying degrees

Internal vs. External Knowledge

Query: “Ricky Hatton Cause of Death”

```
Ricky Hatton, the former
professional boxer, is alive.
There might be some confusion
or misinformation, as of the
latest information, he has
not passed away.
```

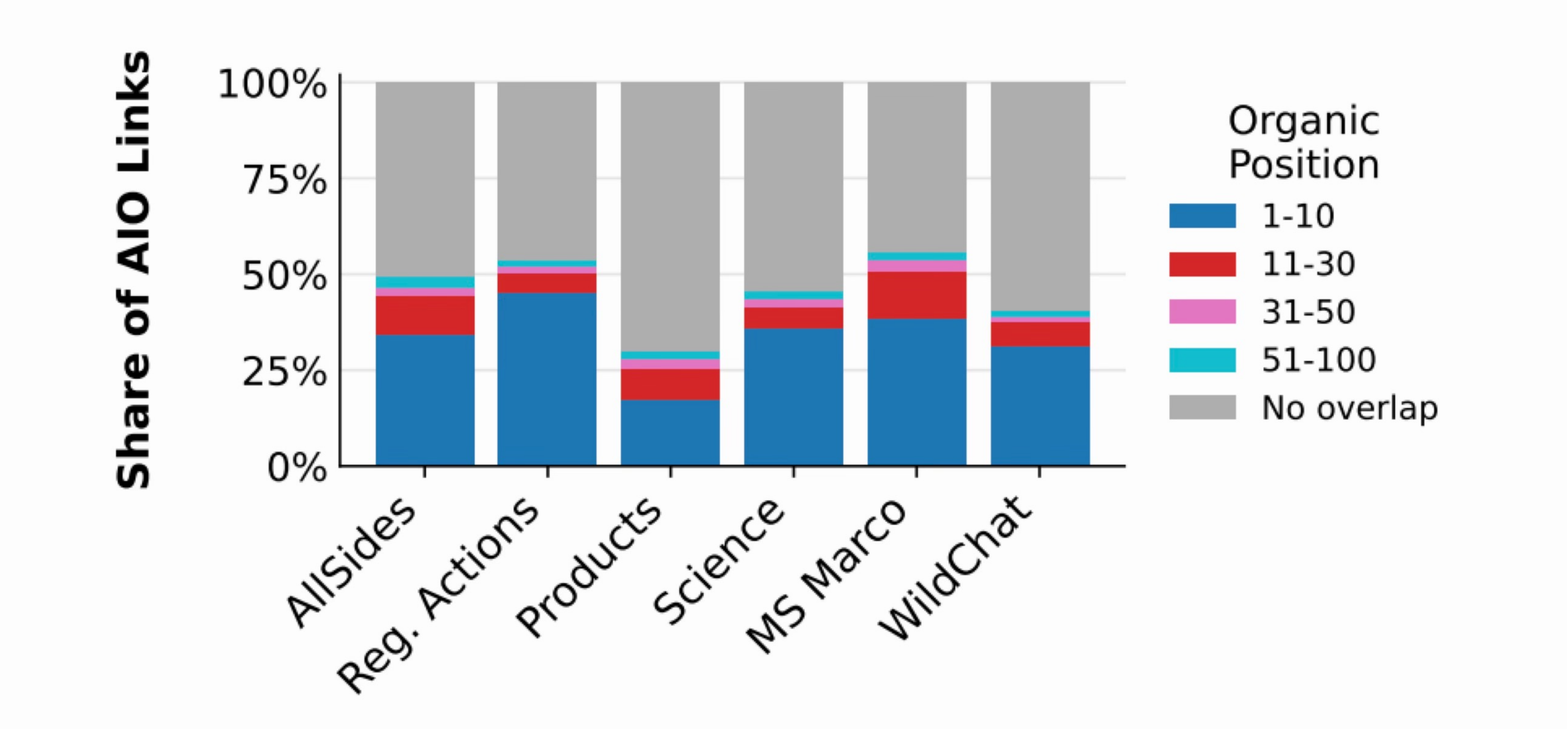
GPT-Tool

```
Former world boxing champion
Ricky Hatton has died at the
age of 46.
His body was found by
Manchester Police[…].
```

Gemini

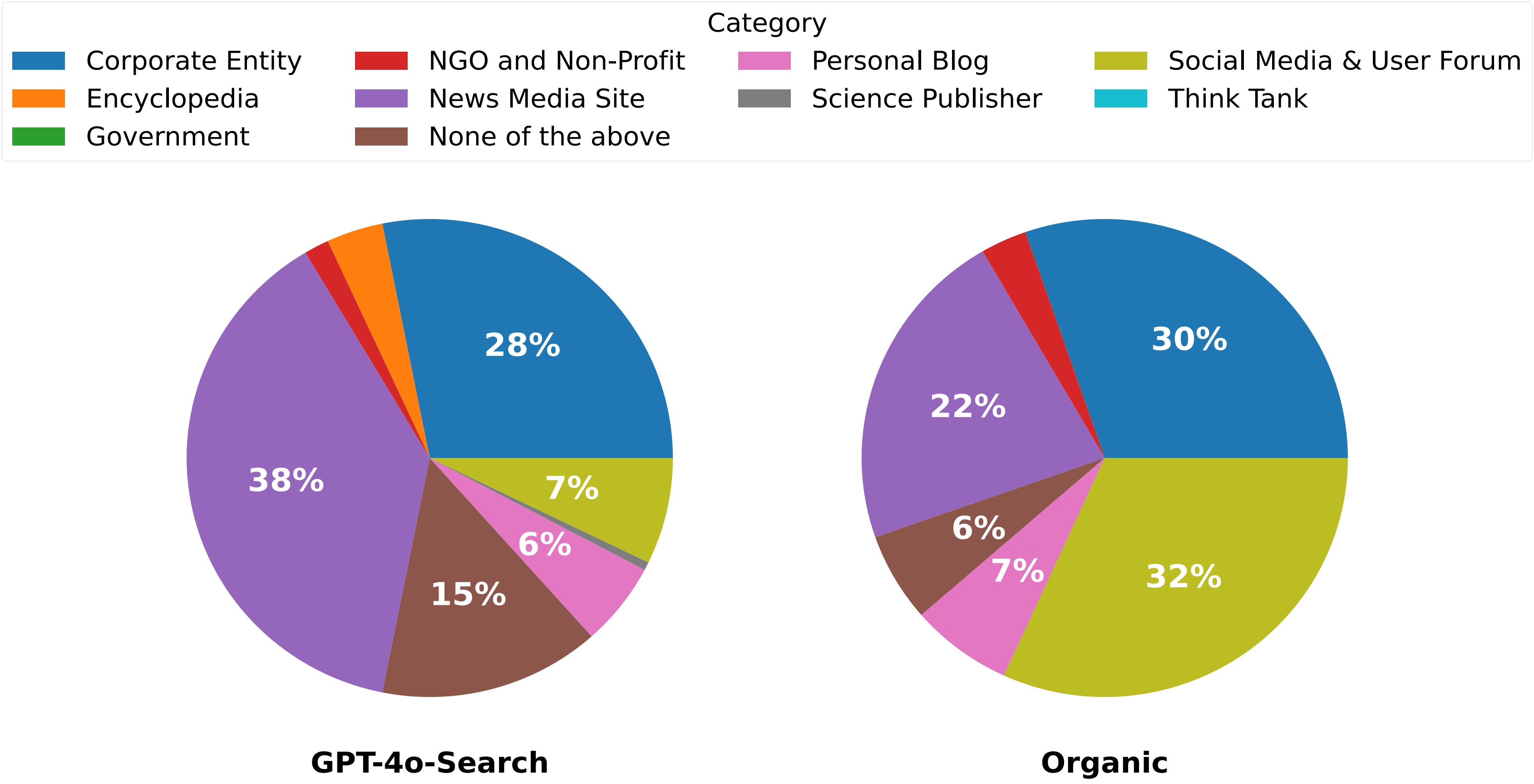
Relying on internal knowledge only can affect the accuracy of results

Which Sources Are Used?



Agents' sources are not overlapping much with traditional search

What Kinds of Sources Are Consulted?



Agents surface different information

Querying Search Agents Repeatedly

Query:

"Does the Soria province have a border with Navarre?"

12:00

No, the Province of Soria does not directly border Navarre[…]

12:05

Yes, (in Castile and León) and the region of Navarre share a border[…]

Engine	**Temp. = 0**	
	$t_0 \rightarrow t_5$	$t_0 \rightarrow t_{24}$
GPT-Tool	9%	10%
GPT-Search	16%	17%
Gemini	15%	15%
AIO	17%	16%
Sonar	27%	28%

% Queries with a decision flip

LLM stochasticity can lead to undesirable effects

From Search Agents to Tool-Using Agents

What is the total cost of 3 nights at €137 per night plus 19% VAT, split between 4 people?

```
Thought: Need exact arithmetic.
Action: calculator("(3 * 137 * 1.19) / 4")
Observation: 122.3775
Final: Each person pays about €122.38.
```

When should the model delegate to an external system?

Millions of Tools Available

Clusters of APIs

- Marketplaces like RapidAPI, MCP servers, etc.

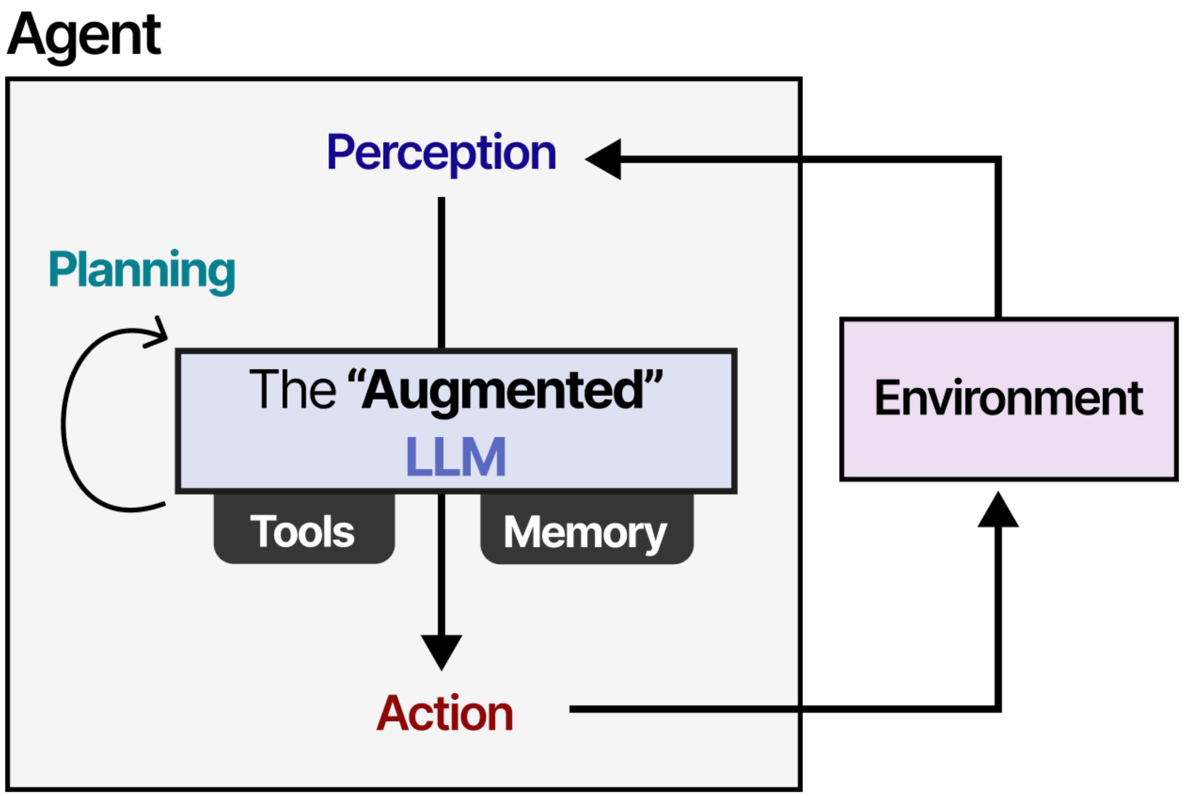
Tool Use ≠ Full Agency

- Tool-augmented LLM:
  - Question → Tool call → Answer

- Agent:
  - Goal → Decide → Act → Observe → Update state
    → Decide → Act → Observe → …
    → Finish

Agency requires a control loop

Agent Harness



ReAct: Reason + Act

ReAct Prompt

ReAct steps

- **Solve the question from the user**

Use the following interleaving steps:

- **Thought**
- **Action**
- **Observation**

Step descriptions

A **thought** can reason about the current situation.

An **action** can be one of two types:
- Calculator(formula)
- Weather()

An **observation** is the result of an action.

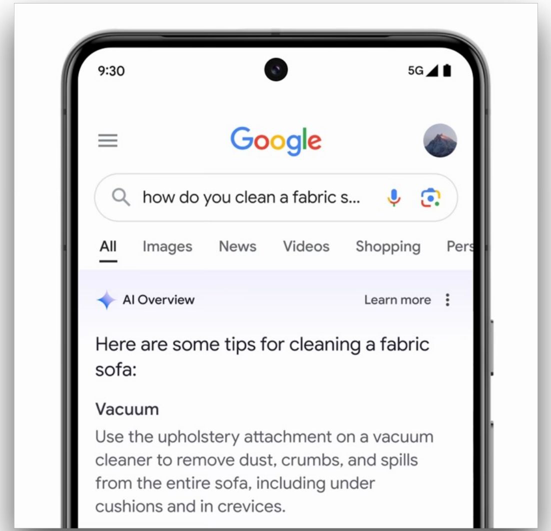
Interleave Reasoning and Acting

ReAct: Reason + Act

```
Thought: What do I need to do?
Action: Use a tool
Observation: What happened?
Thought: What does this imply?
Action: Next tool call
...
Final: Answer / completed task
```

Interleave Reasoning and Acting

Examples of Agents

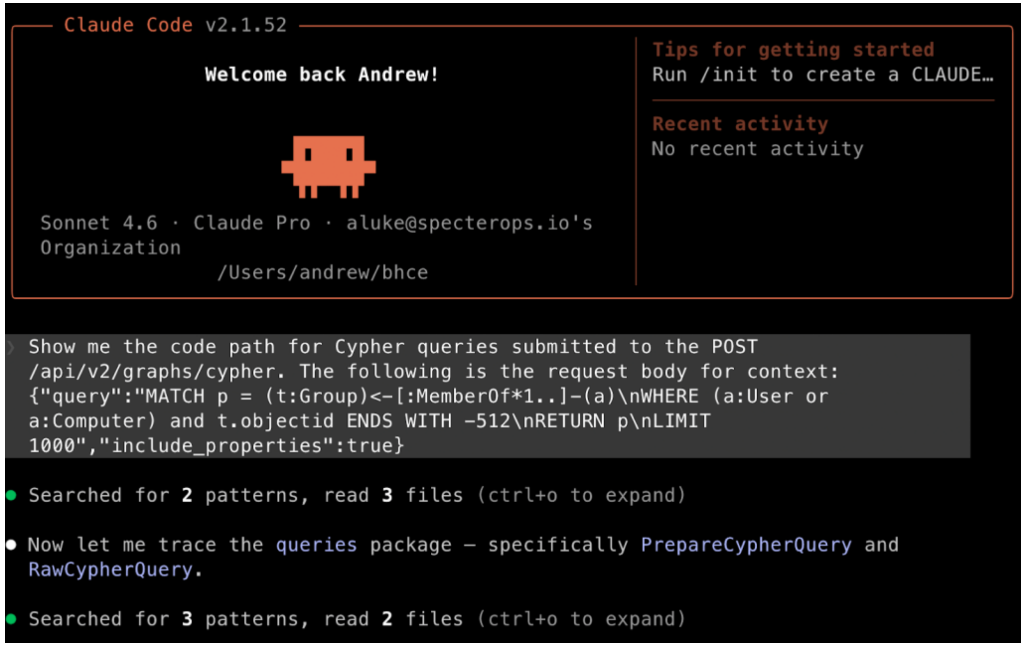


Why do we need agents?

Tools expand what the model can do

- Choose actions
- Interact with environment
- Observe feedback
- Adapt over multiple steps
- Use tools
- Maintain state
- Stop when done
- Ask for approval

Inside a Coding Agent

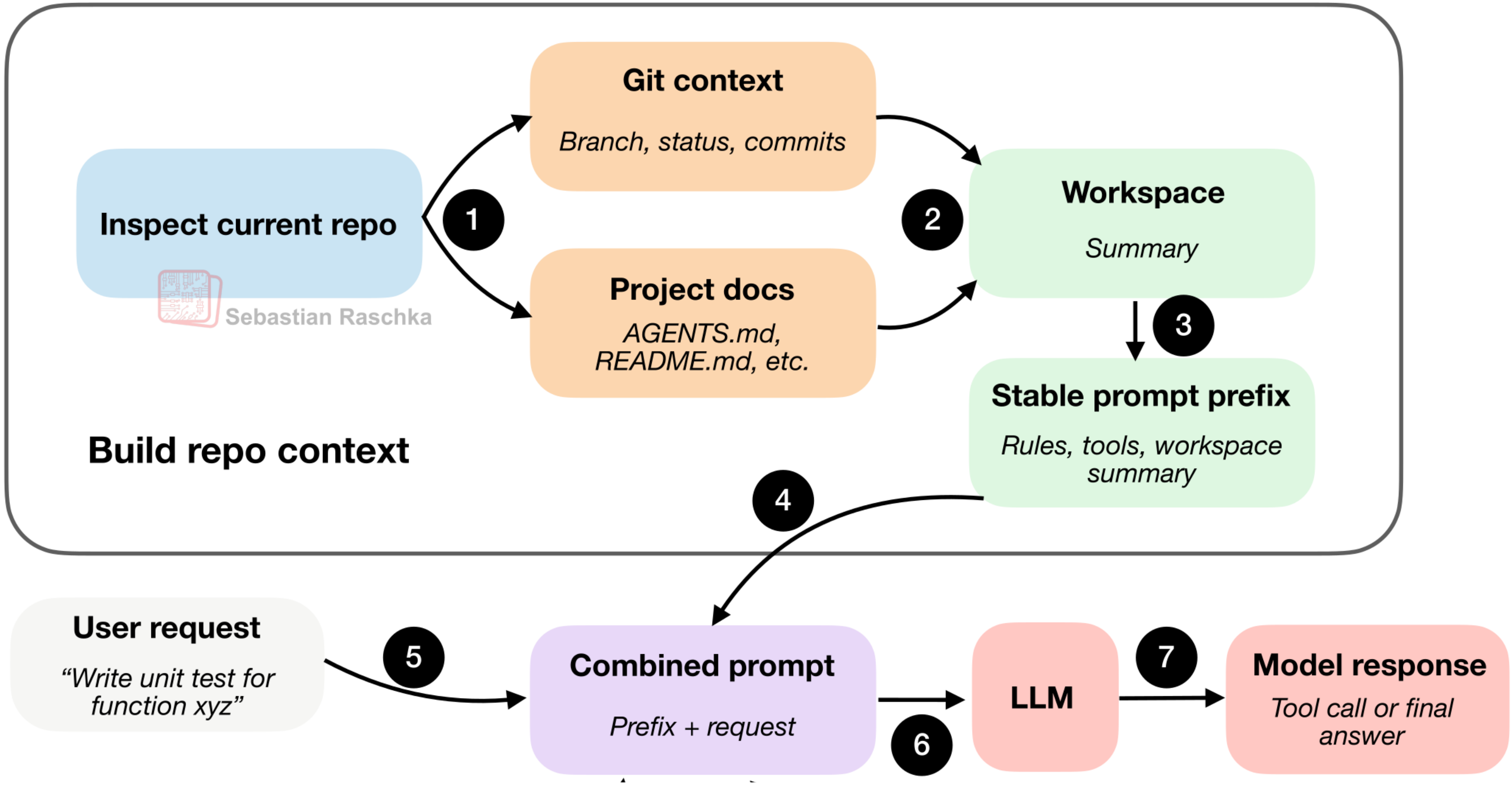


Components of a coding agent:

1. Live repo context
2. Prompt cache
3. Tools
4. Context management
5. Session memory
6. Subagents

Blog Post: Components of A Coding Agent

Inside a Coding Agent



Blog Post: Components of A Coding Agent

Further Reading

- Chain of Thought Paper
- RAG Paper
- ReAct Paper

Next Week:
- Hands-on Agents
- System Design
- MCP Servers

Exercise